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A Household Optimization Problems

In this section we describe the dynamic optimization problem of households who are not employed (the decision problem for employed workers is contained in Section 2 of the main text).

The problem of the passive non-participant (n_0) is

$$v_0(n_0, a_0, z_0) = \max_{\{c_t\}_{t\geq 0}} \mathbb{E}_0^{CD} \int_0^{\mathfrak{t}^{n_1}} e^{-\rho t} \mathfrak{u}^n(c_t, 0)\, dt + e^{-\rho \mathfrak{t}^{n_1}} v_{\mathfrak{t}^{n_1}}(n_1, a_{\mathfrak{t}^{n_1}}, z_{\mathfrak{t}^{n_1}})$$

$$\begin{aligned} s.t.\ c_t + \dot{a}_t &= r_t a_t + \phi_t \\ a_t &\geq 0 \end{aligned} \tag{A1}$$

Passive non-participants do not receive any job opportunity. At rate $\eta^{n_0 n_1}$, with $\mathfrak{t}^{n_1}$ being the first arrival rate of this event, they become active non-participants and enter employment status n_1. When in the passive state, workers only get to choose their consumption flow c_t.

The problem of an unemployed household who is not eligible for UI benefits is:

$$\begin{aligned} v_0(u_0, a_0, z_0) = \max_{\{c_t\}_{t\geq 0}, \mathfrak{t}^*} \mathbb{E}_0^{CD} \Bigg[& \int_0^{\mathfrak{t}^{\min}} e^{-\rho t} \mathfrak{u}^u(c_t, 0)\, dt + \mathbb{I}_{\{\mathfrak{t}^{\min} = \mathfrak{t}^e\}} e^{-\rho \mathfrak{t}^e} v_{\mathfrak{t}^e}(e, a_{\mathfrak{t}^e}, z_{\mathfrak{t}^e}) \\ & + \mathbb{I}_{\{\mathfrak{t}^{\min} = \mathfrak{t}^*\}} e^{-\rho \mathfrak{t}^*} v_{\mathfrak{t}^*}(n_1, a_{\mathfrak{t}^*}, z_{\mathfrak{t}^*}) + \mathbb{I}_{\{\mathfrak{t}^{\min} = \mathfrak{t}^{n_0}\}} e^{-\rho \mathfrak{t}^{n_0}} v_{\mathfrak{t}^{n_0}}(n_0, a_{\mathfrak{t}^{n_0}}, z_{\mathfrak{t}^{n_0}}) \Bigg] \end{aligned}$$

$$\begin{aligned} s.t.\ c_t + \dot{a}_t &= r_t a_t + \phi_t \\ a_t &\geq 0 \end{aligned} \tag{A2}$$

Ineligible unemployed workers receive a job opportunity at rate λ_{zt}^{ue} (with $\mathfrak{t}^e$ being the arrival time of this event) and always find it optimal to take it. At any time $\mathfrak{t}^*$ during the unemployment spell, the individual can quit the labor force $\left(\mathfrak{p}_t^u = 0\right)$. Finally, at rate $\eta^{n_1 n_0}$ (with $\mathfrak{t}^{n_0}$ being the first arrival rate of this shock) they can become passive non-participants.

The problem of an unemployed household who is eligible for UI benefits is:

$$
\begin{aligned}
v_0\left(u_1, a_0, z_0\right) = \max_{\{c_t\}_{t\geq 0}, \mathrm{t}^*} \quad & \mathbb{E}_0^{CD}\Bigg[\int_0^{\mathrm{t}^{\min}} e^{-\rho t}\mathfrak{u}^{u}\left(c_t, 0\right) dt + \mathbb{I}_{\left\{\mathrm{t}^{\min}=\mathrm{t}^{e}\right\}} e^{-\rho \mathrm{t}^{e}} \max\left\{v_{\mathrm{t}^{e}}\left(e, a_{\mathrm{t}^{e}}, z_{\mathrm{t}^{e}}\right), v_{\mathrm{t}^{e}}\left(u_1, a_{\mathrm{t}^{e}}, z_{\mathrm{t}^{e}}\right)\right\} \\
& + \mathbb{I}_{\left\{\mathrm{t}^{\min}=\mathrm{t}^*\right\}} e^{-\rho \mathrm{t}^*} v_{\mathrm{t}^*}\left(n_1, a_{\mathrm{t}^*}, z_{\mathrm{t}^*}\right) + \mathbb{I}_{\left\{\mathrm{t}^{\min}=\mathrm{t}^{u_0}\right\}} e^{-\rho \mathrm{t}^{u_1}} v_{\mathrm{t}^{u_0}}\left(u_0, a_{\mathrm{t}^{u_0}}, z_{\mathrm{t}^{u_0}}\right) \\
& + \mathbb{I}_{\left\{\mathrm{t}^{\min}=\mathrm{t}^{n_0}\right\}} e^{-\rho \mathrm{t}^{n_0}} v_{\mathrm{t}^{n_0}}\left(n_0, a_{\mathrm{t}^{n_0}}, z_{\mathrm{t}^{n_0}}\right)\Bigg] \\
s.t.\ c_t + \dot{a}_t &= r_t a_t + (1-\tau_t) b_t\left(z_t\right) + \phi_t \\
a_t &\geq 0
\end{aligned} \tag{A3}
$$

Besides receiving job opportunities and choosing whether to take them, choosing to drop out of the labor force, and exogenously switching to passive non-participant status, the eligible unemployed could lose its entitlement to UI benefit at rate $\lambda_z^{u_1 u_0}$, with t^{u_0} being the first arrival time of this event.

Finally, the problem of the employed household is:

$$
\begin{aligned}
v_0\left(e, a_0, z_0\right) = \max_{\{c_t\}_{t\geq 0}, \mathrm{t}^*} \quad & \mathbb{E}_0^{CD}\Bigg[\int_0^{\mathrm{t}^{\min}} e^{-\rho t}\mathfrak{u}^{e}\left(c_t, h_t; z_t\right) dt + \mathbb{I}_{\left\{\mathrm{t}^{\min}=\mathrm{t}^{u}\right\}} e^{-\rho \mathrm{t}^{u}} v_{\mathrm{t}^{u}}\left(u_1, a_{\mathrm{t}^{u}}, z_{\mathrm{t}^{u}}\right) \\
& + \mathbb{I}_{\left\{\mathrm{t}^{\min}=\mathrm{t}^*\right\}} e^{-\rho \mathrm{t}^*} v_{\mathrm{t}^*}\left(n_1, a_{\mathrm{t}^*}, z_{\mathrm{t}^*}\right) + \mathbb{I}_{\left\{\mathrm{t}^{\min}=\mathrm{t}^{n_0}\right\}} e^{-\rho \mathrm{t}^{n_0}} v_{\mathrm{t}^{n_0}}\left(n_0, a_{\mathrm{t}^{n_0}}, z_{\mathrm{t}^{n_0}}\right)\Bigg] \\
s.t.\ c_t + \dot{a}_t &= r_t a_t + (1-\tau_t) w_t z_t h_t + \phi_t \\
a_t &\geq 0
\end{aligned} \tag{A4}
$$

Employed workers (e) can be laid-off at rate λ_{zt}^{eu}, in which case they become eligible for UI benefits ($u = u_1$). Let t^{u} be the first arrival time of this Poisson shock. At every instant t^*, the employed worker can choose to quit the labor force ($\mathfrak{p}_t^e = 0$).[1] In addition, an employed worker can exogenously switch to passive non-participant status at rate η^{en_0}, with t^{n_0} being the first arrival time of this event.

[1] Quitting into unemployment is never optimal, because the worker would not receive UI benefits, and would pay a higher disutility cost κ^u for the opportunity to be re-employed at the same wage.

B Intermediate goods sector with capacity constraints

The problem of the intermediate firm follows the description in the main text.[2] In keeping with the other symmetry assumptions made in this block of the model, we assume that intermediate firms are also subject to a common and exogenous maximum capacity constraint $y_{jt} \leq \bar{Y}_t$. Following Comin et al. (2023), the problem of the intermediate firm can be written as

$$\max_{\{p_{jt}\}_{t\geq 0}} \int_0^\infty e^{-(\rho+\chi)t}\left[\frac{p_{jt}}{P_t}y_{jt} - (1-\tau_\ell)w_t\ell_t - \frac{\Theta}{2}\left(\frac{\dot{p}_{jt}}{p_{jt}} - \pi^*\right)^2 Y_t\right] \tag{B1}$$

$$s.t.$$

$$y_{jt} = \Gamma\ell_{jt}^{(1-\alpha)}, \qquad y_{jt} = \left(\frac{p_{jt}}{P_t}\right)^{-\epsilon} Y_t, \qquad y_{jt} - \bar{Y}_t \leq 0 \tag{B2}$$

After building the Hamiltonian, computing the necessary conditions for optimality, and dropping the j index because of symmetry, we obtain

$$\lambda_t = \frac{\Theta(\pi_t - \pi^*)Y_t}{p_t}$$

$$(\rho+\chi)\lambda_t - \dot{\lambda}_t = -\left(\frac{p_t}{P_t} - (1-\tau_\ell)m_t - \mu_t\right)\epsilon\left(\frac{p_t}{P_t}\right)^{-\epsilon-1}\frac{Y_t}{P_t} + \left(\frac{p_t}{P_t}\right)^{-\epsilon}\frac{Y_t}{P_t}$$

$$\mu_t(y_t - \bar{Y}_t) = 0$$

where $m_t = w_t/\left((1-\alpha)\Gamma\ell_t^{-\alpha}\right)$ denotes the firms' real marginal cost, and μ_t denotes the Lagrange multiplier associated with firm's capacity constraint. The last equation denotes the complementarity slackness condition that the multiplier is positive whenever firms hit their capacity constraint.

In a symmetric equilibrium, the optimal price setting decisions of firms yields the price Phillips curve

$$(\rho+\chi)(\pi_t - \pi^*) - \dot{\pi}_t = \frac{\varepsilon}{\Theta}\left((1-\tau_\ell)m_t + \mu_t - \frac{\epsilon-1}{\epsilon}\right) \tag{B3}$$

with the multiplier μ entering the price-setting decision next to the marginal cost.

When the constraint is not binding, firms set their prices optimally to trade off the adjustment cost with their desire to maintain markups at their desired level. When capacity constraints bind—either because demand for a firm's output surges or because its productive capacity is curtailed by input disruptions—firms are instead forced to "price to demand", so that the demand for their product does not exceed their maximum capacity and their production exhausts demand at that price. Note that it would not be optimal

[2] For simplicity, the derivation sets the indexation parameter γ to zero.

for firms to price so their demand is below productive capacity. Moving the multiplier μ_t outside of the brackets and relabelling the resulting term as $\zeta_t^{cc} \equiv \frac{\varepsilon}{\Theta}\mu_t$, we obtain

$$\rho(\pi_t - \pi^*) - \dot{\pi}_t = \frac{\varepsilon}{\Theta}\left((1-\tau_\ell)m - \frac{\epsilon - 1}{\epsilon}\right) + \zeta_t^{cc} \tag{B4}$$

which illustrates that binding capacity constraints $\zeta_t^{cc} > 0$ are isomorphic to positive cost-push shocks ζ_t^p in the price Phillips curve (13).

C Equilibrium

We now formally state the definition of a perfect foresight equilibrium for our economy. A stationary equilibrium (or steady-state) is a particular case of our definition where all decisions, prices, aggregate variables, and distributions are constant over time.

For any given initial distribution of households $\mu_0(s,a,z)$, with $s \in \{e,u,n\}$, and a time path for shocks $\{\zeta_t^k\}_{t\geq 0}$, with $k \in \{d,\ell,m,\bar{Y}\}$, a *perfect foresight equilibrium* is defined as time paths for household consumption, participation and job offer acceptance decisions $\{c_t(s,a,z), \mathfrak{p}_t(s,a,z), \mathfrak{j}_t(s,a,z)\}_{t\geq 0}$, unions' nominal wage setting $\{\omega_{kt}\}_{t\geq 0}$, intermediate producers' hiring decisions $\{\ell_{kt}\}_{t\geq 0}$, real rates of return on real and nominal government bonds $\left\{r_t^b, r_t^m\right\}_{t\geq 0}$, firms' share price $\left\{q_t\right\}_{t\geq 0}$, fiscal variables (UI benefits, taxes, transfer, and debt) $\left\{b_t(z), \tau_t, \phi_t, B_t^g\right\}_{t\geq 0}$, nominal interest rates $\{i_t\}_{t\geq 0}$, aggregate output, consumption, hours worked, price and wage inflation $\left\{Y_t, C_t, h_t, \pi_t, \pi_t^\omega\right\}_{t\geq 0}$, and measures of households $\left\{\mu_t(s,a,z)\right\}_{t>0}$ such that at every t: (i) households optimize; (ii) firms optimize and inflation satisfies the price Phillips curve; (iii) unions optimize and wage inflation satisfies the wage Phillips curve; (iv) no-arbitrage condition holds for all assets; (v) the government budget constraint holds; (vi) the fiscal and monetary authorities follow their policy rules; (vii) the sequence of distributions satisfies aggregate consistency conditions; (viii) asset and annuity markets clear, and (ix) all goods markets clear.

The asset market clears when households' asset holdings equals the equity value plus government real bonds

$$\int a_t d\mu_t = q_t + B_t^g.$$

This, together with the no-arbitrage conditions, determine firm share prices and real interest rates. Market clearing in the continuum of intermediate goods' markets requires $Y_t = y_{jt}$, for all j. Given the set of employed workers, average hours per worker h_t is determined by the demand of intermediate goods producers, $y_{jt} = \Gamma\left(h_t \int_{s_{it}=e} z_{it}\right)^{1-\alpha} di$, for all j. The final good market, $Y_t = C_t + G_t$, clears by Walras law. Lastly, we note that the labor market is frictional and the evolution of the distribution of employed, unemployed, and non participants is implicit in the KFEs for μ_t.

D Parameters

Steady State		
Parameter		Value
Demographics and Preferences		
Death rate	ϱ	$1/432$
Effective discount rate	ρ	0.0053
Labor supply elasticity	σ	1.00
Utility weight on hours	ψ	1.00
Disutility of nonparticipation	κ^n	0
Disutility of searching	κ^u	0.33
Disutility of working	κ^e	1.05
Skill dynamics		
Mean of initial skill distribution	$\bar{z}_0$	$\log(0.68) - \sigma_{z_0}^2/2$
S.D. of initial skill distribution	σ_{z_0}	0.50
Skill mean reversion	γ_z	0.0017
Skill drift while employed	δ_z^+	0.0024
Skill drift while non-employed	δ_z^-	0.0214
Skill diffusion coefficient	σ_z	0.0467
Labor market frictions		
Job-separation rate out of E	λ_z^{eu}	—
Job-finding rate out of U	λ_z^{ue}	—
Job-finding rate out of N	λ_z^{ne}	—
Passive nonparticipation rate during E	η^{en_0}	0.0075
Passive nonparticipation rate during U/N	$\eta^{un_0}, \eta^{n_1 n_0}$	0.0750
Passive nonparticipation exit rate	$\eta^{n_0 n_1}$	0.2500
Technology		
Liquidity premium	ι	0.001
Elast. of subst. between goods	ν	10
Elast. of subst. between tasks	ε	10
Firm productivity	Γ	1.56
Decreasing returns to scale	α	0.20
Fixed costs	Ω	0.30
Taxes, transfers and expenditures		
Government debt	B^g	1.69
UI replacement rate	$\bar{b}$	0.50
UI expiration rate	$\lambda^{u_1 u_0}$	0.167
Labor income tax rate	τ	0.20
Lump-sum transfer	ϕ	0.068
Government expenditures	G	0.17
Intermediate firms labor subsidy	τ^ℓ	0.10
Intermediate firms net subsidy	Ψ^ℓ	0.001

Out of Steady State		
Parameter		Value
Monetary and Fiscal Policy		
Trend inflation	π^*	$0.02/12$
Taylor rule persistence	β_i	0.07
Taylor rule reaction to inflation	β_π	2.25
Taylor rule reaction to unemployment rate	β_u	-0.10
Labor tax response to debt	β_τ	0.008
Phillips Curve		
Indexation of current to past inflation	γ	0.950
Slope of the price Phillips curve	θ	0.050
Slope of the wage Phillips curve	θ_w	0.005
Aggregate Fluctuations		
Cognitive discounting	χ	$-1/3 \log(0.65)$
Elasticity of (λ_z^{eu}) to hours	ϑ^{eu}	-11.00
Elasticity of $(\lambda_z^{ue}, \lambda_z^{ne})$ to hours	$\vartheta^{ue}, \vartheta^{ne}$	24.00

Table D1: Model parameter values expressed in monthly frequency. See Section 4 in the main text for a discussion of parameter choices and targets.

E Impulse Response Functions

We model all shocks as Ornstein-Uhlenbeck diffusion processes

$$d\zeta_t^k = -\gamma_k \zeta_t^k dt + \sigma_k d\mathcal{W}_t^k, \tag{E1}$$

where $d\mathcal{W}_t^k$ is a standard Wiener process. Figure E1 summarizes the IRFs of some key equilibrium aggregate outcomes.

Demand shock. A negative shock to the demand for final goods reduces labor demand, thus hours decline and unemployment rises. In response to weaker labor demand, unions lower nominal wages, which pushes down prices. Output, hours, and inflation comove positively under this shock, which is why we refer to it as a demand shock.[3] The monetary authority responds to these fluctuations by cutting nominal (and real) rates, which helps cushion the decline in consumption expenditures. Labor productivity jumps at impact because of selection into employment. However, approximately 10 quarters later, productivity turns negative as workers who experienced non-employment spells reenter the labor force after suffering some degree of skill losses.

Labor supply shock. The labor supply shock primarily leads to a decline in labor force participation. This reduction to labor supply creates a labor shortage, which in turn leads to an increase in hours per worker and a reduction in unemployment. The tight labor market exerts mild upward pressure on inflation, prompting the monetary authority to raise interest rates. The shock has also a pronounced impact on labor productivity through selection, as it is predominantly lower-skilled workers who exit the labor force.

Cost-push shock. As discussed above, it is more intuitive to describe the impact of a cost-push shock in the case of a positive shock to firms' desired markup ($\zeta_t^m > 0$). The shock induces firms to raise prices given their real marginal cost, putting upward pressure on inflation and causing real wages to fall. In response, the monetary authority increases nominal (and real) interest rates to contain inflation. The resulting tightening in monetary policy reduces aggregate demand, leading to lower hours worked and higher unemployment. Overall, this shock generates a negative comovement between inflation and output.

[3] This disturbance is akin to a "risk-premium" shock, common in the literature on estimated New Keynesian DSGE models, and routinely found to contribute significantly to the overall fluctuations of total hours worked in the US economy (see, e.g. Smets and Wouters, 2007; Galí et al., 2012).

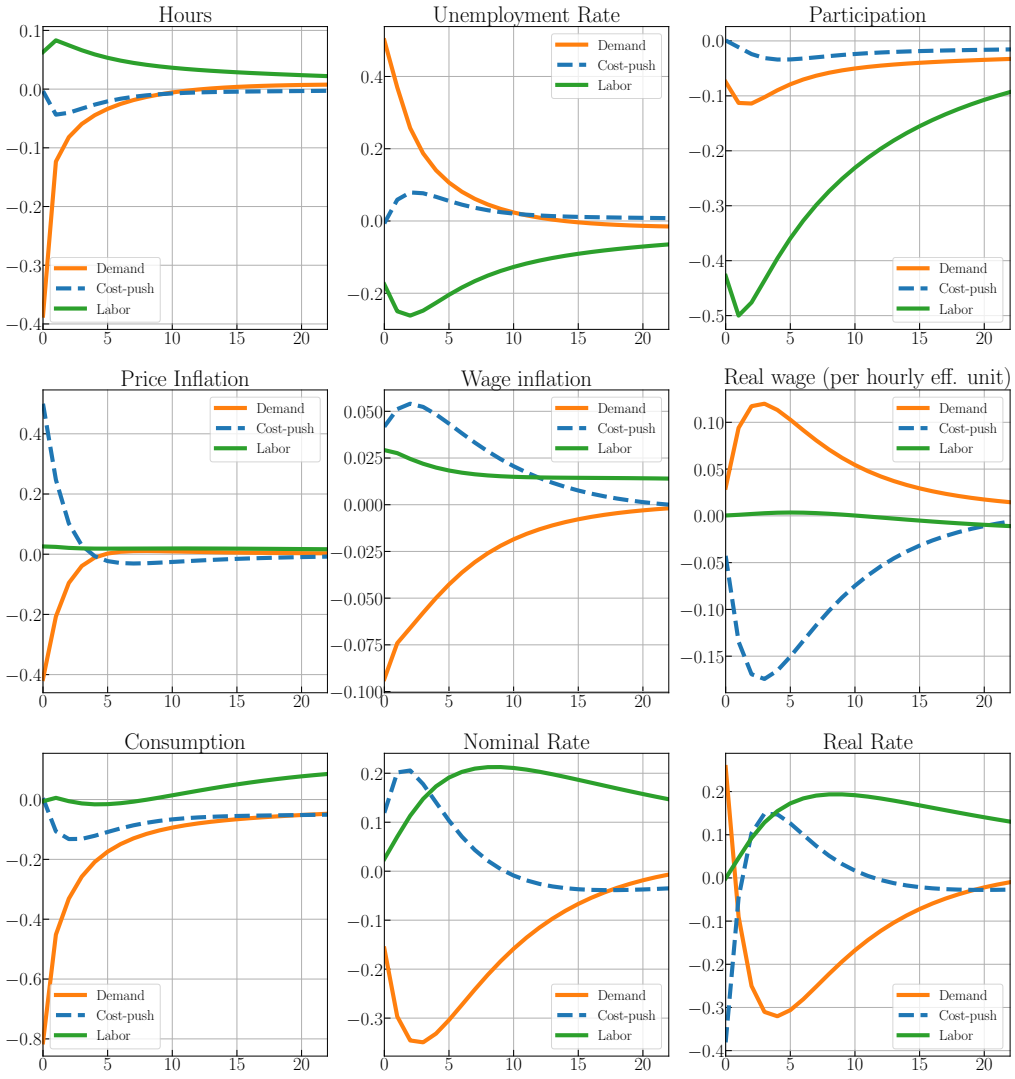


Figure E1: Impulse response functions of some of model's aggregate variables to demand (ζ^d), mark-up (ζ^m), and labor supply (ζ^ℓ) shocks.

F Additional Results on Welfare Analysis

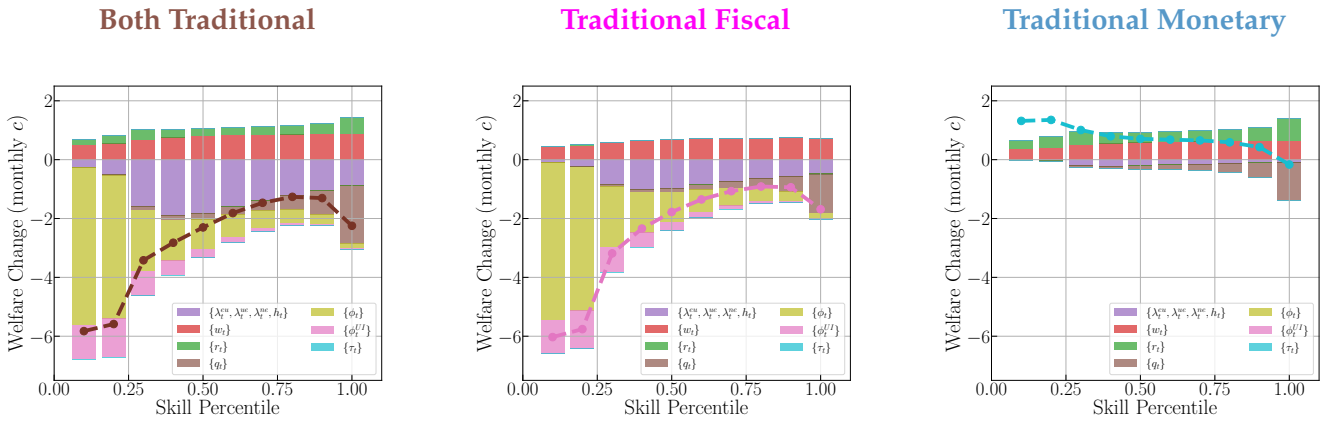


Figure F1: Full decomposition of welfare gains of counterfactual policies into all the individual components determining the labor market channel, the financial return channel and the transfers and taxes channel.

Note: Welfare effects are computed using the money-metric (18) and are expressed as a share of steady-state monthly consumption. Columns correspond to different counterfactual scenarios. *Note:* Values are expressed as a share of steady-state monthly consumption. Columns correspond to different counterfactual scenarios. The first row reports the distribution of welfare changes across households. The second row reports mean welfare changes by skill decile: the dashed line shows the total effect, and the bars decompose it into the effect of different inputs to the household problem (see main text for channel definitions). Differences between the total effect and the sum of the channels reflect non-linearities in the value function.

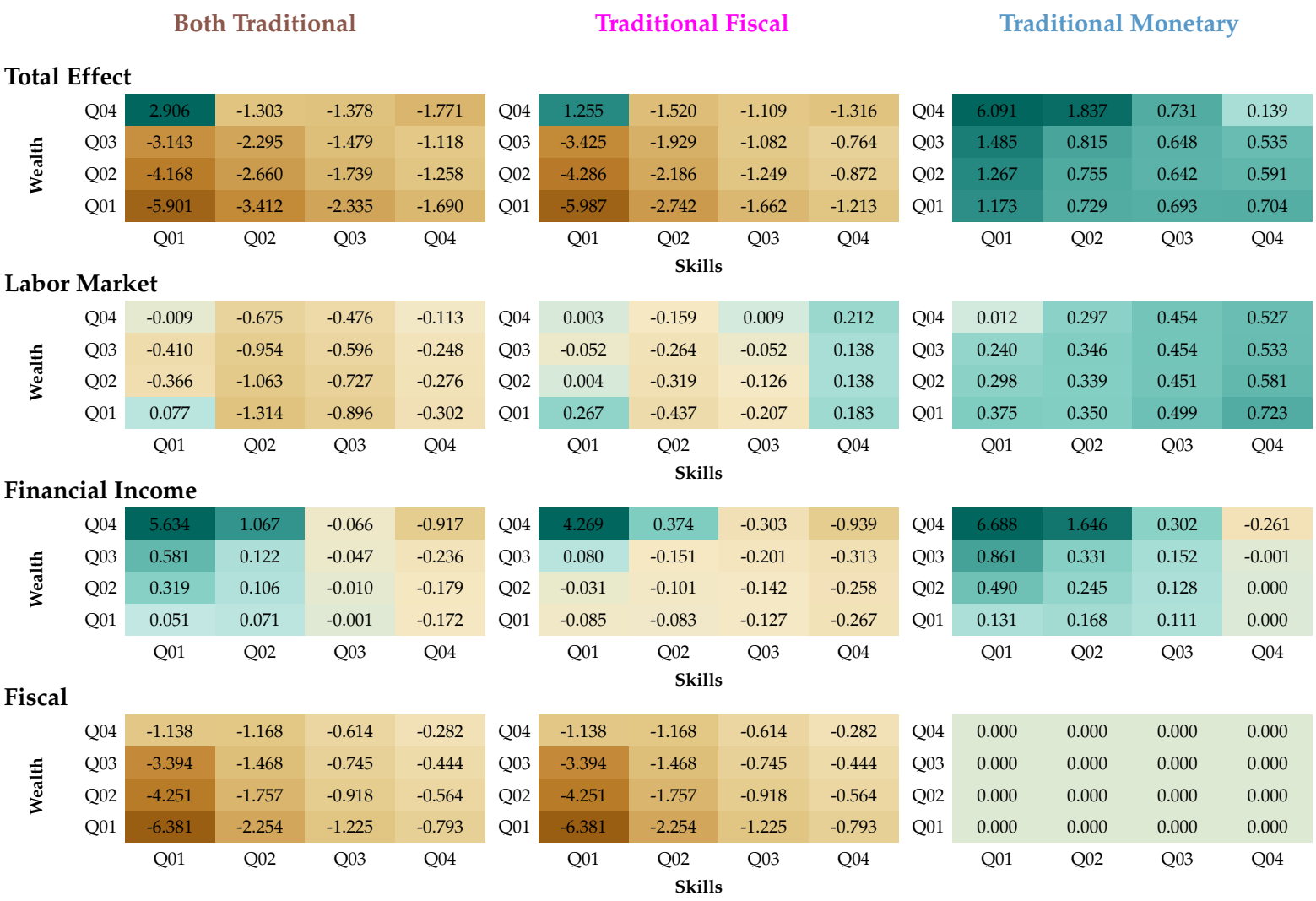


Table F1: Welfare gains of counterfactual policies by skill and wealth.

Note: Welfare effects are computed using the money-metric (18) and are expressed as a share of steady-state monthly consumption. Columns correspond to different counterfactual scenarios. The top row displays the total welfare effects, while the second to fourth rows decompose the effect into the labor market, financial returns, and transfer and taxes channels (see main text for a description of the channels). Measures over skill and wealth groups are calculated by averaging our welfare metric according to the joint distribution of skills and wealth in steady state.

References

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